MILITARY SPECIFICATION

MIXER, TRANSPONDER SUPPRESSION TD-937()/SPX

This specification is approved for use by all Departments and Agencies of the Department of Defense.

1. SCOPE

1.1 This specification covers a Transponder Suppression Mixer which provides the necessary suppression gates to a single transponder from up to six suppression sources.

2. APPLICABLE DOCUMENTS

2.1 The following documents of the issue in effect on date of invitation for bids or request for proposal, form a part of the specification to the extent specified herein.

SPECIFICATIONS

MILITARY

MIL-S-901	-	Shock Tests, H.I. (HIGH-IMPACT), Shipboard Machinery, Equipment and Systems, Requirements for (Navy)
MIL-Q-9858	-	Quality Program Requirements
MIL-E-16400	-	Electronic Equipment, Naval Ship and Shore, General Specification (Navy)
MIL-E-17555	-	Electronic and Electrical Equipment, Accessories, and Repair Parts, Packaging and Packing of
MIL-C-39012/21	-	Connector, Coaxial, Radio Frequency, (Series Bnc-receptacle-jam Nut Mounted-class 1i)

STANDARDS

MILITARY

MIL-STD-105	-	Sampling procedures and Tables for Inspection by Attributes
MIL-STD-108	-	Definition of and Basic Requirement for Enclosure for Electric and Electronic Equipment.
MIL-STD-167	-	Mechanical Vibrations of Shipboard Equipment
MIL-STD-275	-	Printed Wiring for Electronic Equipment
MIL-STD-454	-	Standard General Requirements for Electronic Equipment
MIL-STD-461	-	Electromagnetic Interference Characteristics Requirements for Equipment
MIL-STD-470	-	Maintainability Program Requirements (for Systems and Equipments)
MIL-STD-471	-	Maintainability Demonstration
MIL-STD-781	-	Reliability Tests Exponential Distribution
MIL-STD-785	-	Reliability Program for Systems and Equipment Development and Production

PUBLICATIONS

MILITARY

MIL-HDBK-217	-	Reliability Stress and Failure Rate Data for Electronic Equipment
MIL-HDBK-472	-	Maintainability Prediction

(Copies of specifications, standards, drawings, and publications required by suppliers in connection with specific procurement functions should be obtained from the procuring activity or as directed by the contracting officer.)

Source: https://assist.dla.mil -- Downloaded: 2026-10-20T14:07Z
Check the source to verify that this is the current version before use.

3. REQUIREMENTS

3.1 General. The equipment shall be in accordance with all the provisions of MIL-E-16400 except as hereinafter specified. The design objectives as specified in MIL-E-16400 for simplicity, ease of maintenance and installation shall be considered during all phases of equipment design.

3.2 First article sample. Prior to beginning production, a sample shall be tested as specified in 4.3 (see 6.2.1).

3.3 Reliability.

3.3.1 Quantitative reliability requirement. The equipment shall be designed for a 20,000 hour mean-time-between-failure (MTBF). For reliability demonstration tests the specified MTBF shall be 6,000 hours.

3.4 Design requirements.

3.4.1 Composition. To the maximum extent practicable, the equipment shall consist of solid-state circuitry mounted on printed circuit cards, mounted in an enclosure which is drip-proof in accordance with MIL-STD-108.

3.4.1.1 Printed circuit cards. All printed circuit cards used shall be designed and manufactured in accordance with MIL-STD-275.

3.4.2 Environmental conditions. The equipment shall be capable of operation under any reasonable combination of the following conditions as specified in MIL-E-16400.

(a) Temperature-class 2
(b) Humidity, up to 95 percent relative.
(c) Shock, vibration, inclination and noise

3.4.3 Electrical design.

3.4.3.1 Primary power line voltage and frequency. The equipment shall be capable of operating from a power source whose characteristics are within the following ranges:

(a) Voltage: 115 VAC ±10 percent single phase
(b) Frequency: 50,60, and 400 Hz ±10 percent

3.4.3.1.1 Overload-protection. Fuses, if used, shall be provided with blown-fuse indicators. The use of circuit breakers, selected in accordance with MIL-E-16400, is preferred.

3.4.3.2 External connectors. All signal connectors shall be type C, and all power and control connectors shall be chosen from the AN or MS series. All connectors shall be on the rear panel.

3.4.3.2.1 DC isolation. The video signal inputs shall be capacitively coupled to insure proper system operation with signals which have a dc reference other than ground. This requirement shall not be waived unless the contractor demonstrates to the command or agency concerned and obtains approval of a design technique which is immune to such input signals. This requirement applies to channels A and B only.

3.4.3.2.2 Output reference level. The video signal outputs shall be referenced to ground -0.1 volt to +0.2 volt except as hereinafter specified.

3.4.3.3 Electromagnetic interference emission, susceptibility, and equipment and systems self-compatibility. The equipment shall operate within the electromagnetic interference emission, susceptibility, and equipment and systems self-compatibility limits in accordance with class Gp of MIL-STD-461.

3.4.3.4 Electrical design. The following data will be submitted to the command or agency concerned for approval.

(a) Schematic diagrams of each electrical circuit card and subassembly.
(b) Test specification for each subassembly.

3.4.3.5 Synchros. Synchros are not required.

Source: https://assist.dla.mil -- Downloaded: 2026-10-20T14:07Z
Check the source to verify that this is the current version before use.

3.4.4 Maintenance.

3.4.4.1 Test points and test features are required, and shall comply with the following:

3.4.4.1.1 External test points. External and internal test points shall be provided on the front panel to allow malfunction isolation to the defective equipment. Location of, and parameters to be measured will be subject to the approval of the command or agency concerned. Mechanical design of the front panel test points shall be as specified herein. It shall be possible to monitor the six input signals and the suppression output at the front panel test points.

3.4.4.1.2 Internal test points. Internal test points shall be provided on each circuit board or module. Observation of these test points will identify a defective board or module immediately. All test points for printed circuit cards shall be brought to the top of the card. The test points shall be of the probe type, suitable for insertion of a standard, flexible tip probe.

3.4.4.2 Maintainability.

3.4.4.2.1 Quantitative maintainability requirements. The equipment shall have an equipment repair time (ERT) of fifteen (15) minutes when corrective maintenance by module or circuit card is performed.

3.4.4.3 Service and access. The equipment design shall permit easy and ready access to internal parts, test points, and adjustments. Removal and replacement of subassemblies shall not require the removal or displacement of other subassemblies.

3.4.4.3.1 Accessibility. Equipment procured under this specification shall be designed to permit ready access to internal test points and replaceable modules via the front of the equipment. Requirement 36 of MIL-STD-454 shall apply.

3.4.4.3.1.1 Loosening of captive fasteners on the front panel shall provide for withdrawal of the equipment, from the case, for maintenance.

3.4.4.3.1.2 Slides/guides with safety stops to prevent accidental complete removal shall be employed. Safety interlocks and by-pass switches, selected in accordance with MIL-E-16400 and MIL-STD-454, are required.

3.4.4.3.1.3 The internal power and signal cables shall be of sufficient length to allow full extension of the unit from the case. Provisions must be supplied to support all cables coming from the rear panel to the removable component chassis, such that these cables sustain no damage.

3.4.4.3.2 Accessibility of external test points. Test points shall be provided on the front panel of the equipment. Signals available shall be all of the equipment input/output signals. Type BNC UG-625 connectors in accordance with MIL-C-39012/21 shall be mounted on the front panel of the equipment for measurement of these signals.

3.4.4.4 Reversibility. The equipment shall incorporate features such that it is mechanically or electrically impossible to install incorrectly its modules, circuit card assemblies, and subassemblies.

3.4.5 Mechanical design. The mechanical design of the equipment shall conform to the requirements of the following sub-paragraphs:

3.4.5.1 Shock. The equipment shall be designed in accordance with the requirements of Grade A, Class I, Type A, on the lightweight machine, in accordance with MIL-S-901.

3.4.5.2 Vibration. Vibration resistance shall be in accordance with the requirements of type I as defined by MIL-STD-167.

3.4.5.3 Enclosure. The degree of enclosure shall be dripproof, in accordance with the requirements of MIL-STD-108.

3.4.6 Controls, indicators, and panel layout.

3.4.6.1 Tactile identification of controls is not required.

Source: https://assist.dla.mil -- Downloaded: 2026-10-20T14:07Z
Check the source to verify that this is the current version before use.

3.4.6.2 Classification. All controls shall be classified as continually required except those used primarily in initial installation and adjustment or for maintenance adjustment and alignment.

3.4.7 Supplementary requirements.

3.4.7.1 Instruction plates. Instruction plates are not required.

3.4.7.2 Time meters. Time meters are not required.

3.5 Performance and detail requirements.

3.5.1 Design and construction details.

3.5.1.1 Electrical connectors. The equipment shall be provided with one connector for AC power input. The equipment shall be provided with one Type C coaxial connector for each of the six trigger inputs and the suppression gate output.

3.5.1.1.1 Power connector. The power connector shall be a three pin MS or AN (male) connector to accept 115VAC, 1 ϕ, with the following pin assignments:

pin A 115 VAC
pin B Ground
pin C 115 VAC

3.5.1.2 Front panel controls. The equipment front panel controls shall consist of a Power ON-OFF toggle switch.

3.5.1.3 Front panel indicators. The equipment shall be provided with a Power - On indicator lamp.

3.5.1.4 Size. The dimensions of the equipment shall not exceed the following:

(a) Height - 6 inches
(b) Width - 5 inches
(c) Depth - 13 inches

The envelope described by these dimensions includes all projections from the surfaces of the equipment such as connectors, handles, and screw heads.

3.5.1.5 Weight. The weight of the equipment shall not exceed 12 pounds.

3.5.1.6 Mounting. The equipment shall be designed for mounting to a flat surface without resilient mounts.

3.5.1.7 Component chassis. The chassis assembly shall be composed of a front panel and a component chassis. All circuit boards and subassemblies shall be accessible when the chassis is withdrawn from the equipment case. Accessibility shall be as specified in MIL-E-16400 and further described herein. Access to the equipment interior shall be accomplished by releasing no more than six (6) securing devices located on the front panel surface only.

3.5.1.8 Equipment case. The case shall form the top, bottom, and two lateral sides of the rectangular box comprising the equipment enclosure. The case shall have provisions for direct mounting to an equipment shelf by the bottom surface only.

3.5.1.9 Rear panel. The rear panel of the equipment shall be a removable part of the enclosure and not a part of the equipment chassis. All signal input and output connectors shall be installed on the removable rear panel. Test signal outputs shall be on the front panel (see 3.4.4.3.2).

Source: https://assist.dla.mil -- Downloaded: 2026-10-20T14:07Z
Check the source to verify that this is the current version before use.

3.5.1.10 Enclosure. The equipment shall be mounted by devices secured to the base of the enclosure. The equipment shall be designed to permit ready access to internal test points and replaceable modules via the front of the equipment. MIL-STD-454, requirement 36 shall apply as modified below.

3.5.2 Performance.

3.5.2.1 Input signals. The equipment shall operate with input signals having the following characteristics:

(a) Polarity: Positive
(b) Amplitude: 5 to 55 volts
(c) Duration: 0.30 microseconds minimum, 500 microseconds maximum
(d) Rise time: Up to 25 percent of pulse duration, or 0.04 microsecond per volt times pulse amplitude, whichever is smaller.
(e) Fall time: 1 to 5 times pulse rise time
(f) Source impedance 75 ohms resitive, ±5 ohms.

3.5.2.1.1 Any number of the six trigger/gate inputs shall be able to feed any of the three operational channels within the equipment. Selection of the six inputs may be made by link on a printed circuit board as an initial setup procedure. It shall be possible to change the input selection by changing the links.

3.5.2.1.2 The input signals may have up to three volts of noise which shall not affect equipment operation.

3.5.2.2 Input power. The equipment shall require no more than 25 volt-amperes primary power. The equipments power factor shall be no less than 0.85.

3.5.2.3 Output. The equipment shall produce a single output signal in accordance with the characteristics specified as follows:

3.5.2.3.1 Internal channels. The equipment shall contain three channels of operation, the outputs of which shall be mixed so as to produce a single output from the equipment. The channels are identified as A, B, and C. All measurements of equipment performance shall be made at the single equipment output.

3.5.2.3.2 Channels A and B within the equipment shall accept signals from Category A suppression sources, and shall perform in essentially an identical manner. When supplied with an input signal in accordance with 3.5.2.1, the output signal shall be a gate internally adjustable in length over the range of 5 to 50 microseconds for Channel A and 50 to 500 microseconds for Channel B. The gate shall be of a renewable type so that any subsequent trigger will reinitiate the timing cycle with no loss of output. The gate length, once setup, will maintain the output pulse width within ±2 microseconds or ±5 percent, whichever is greater under all service conditions and input pulse conditions. If the input pulse width exceeds the width of the normally adjusted output gate, the output gate from the equipment shall follow that of the input signal within the limits of zero to +2 microseconds. The delay time between the trigger input and the gate output shall not exceed 0.50 microseconds regardless of the input trigger voltage and/or rise time of 3.5.2.1.

3.5.2.3.3 Channel C of the equipment shall regenerate the input (Category B) pulse. The output from the equipment shall follow that of the input within the limits of zero to +2 microseconds. The delay time between input and output shall not exceed 0.50 microseconds regardless of the input trigger voltage and rise time of 3.5.2.1.

3.5.2.3.4 Output characteristics. Unless otherwise stated the output from the equipment shall possess the following characteristics:

(a) Polarity: Positive
(b) Amplitude: 20 ±5 volts
(c) Noise Level: less than 0.5 volts

Source: https://assist.dla.mil -- Downloaded: 2026-10-20T14:07Z
Check the source to verify that this is the current version before use.

(d) Rise time: less than 0.2 microseconds
(e) Fall time: less than 1.0 microseconds
(f) Termination Impedance: 75 ±5 ohms

4. QUALITY ASSURANCE PROVISIONS

4.1 Responsibility for inspection. Unless otherwise specified in the contract or purchase order, the supplier is responsible for the performance of all inspection requirements as specified herein. Except as otherwise specified in the contract or order, the supplier may use his own or any other facilities suitable for the performance of the inspection requirements specified herein, unless disapproved by the Government. The Government reserves the right to perform any of the inspections set forth in the specification where such inspections are deemed necessary to assure supplies and services conform to prescribed requirements.

4.1.1 Government verification. All quality assurance operations performed by the contractor will be subject to Government verification at any time. Verification will consist of (a) surveillance of the operations to determine that practices, methods, and procedures of the written system description are being properly applied, (b) Government product inspection to measure quality of product to be offered for acceptance, and (c) Government product inspection of delivered products to assure compliance with all requirements of this specification. Failure of the contractor to promptly correct deficiencies discovered by him or of which he is notified shall be cause for suspension of acceptance until corrective action has been made or until conformance of product to prescribed criteria has been demonstrated.

4.2 Classification of inspection. The inspection requirements specified herein are classified as follows:

(a) First article inspection (see 4.3)
(b) Quality conformance inspection (see 4.4)
    (1) Production inspection (see 4.4.1)
    (2) Production control inspection (see 4.4.2)
    (3) Environmental tests (see 4.4.3)
(c) Reliability testing (see 4.5)
(d) Maintainability demonstration (see 4.6)
(e) Inspection of preparation for delivery (see 4.7)

4.3 First article inspection. Unless otherwise specified (see 6.2.1) one (1) unit shall be required for first article inspection. First article inspection shall consist of all examination and testing necessary to determine compliance with the requirements of this specification.

4.3.1 The water cooling, reliability, and accelerated life tests of MIL-E-16400 are not required.

4.4 Quality conformance inspection.

4.4.1 Production inspection. Production inspection shall be made on every equipment offered for delivery. The inspection shall comprise such examination and testing as will prove the workmanship and reveal the omissions and errors of the production process such as functional and performance tests at a limited number of points, tests which detect deviations from design, tests of adjustment, and tests which detect hidden defects of material. Production inspection shall consist of the listed production inspection examinations and tests of MIL-E-16400. The operating test will test each electrical function of the equipment.

Source: https://assist.dla.mil -- Downloaded: 2026-10-20T14:07Z
Check the source to verify that this is the current version before use.

4.4.2 Production control inspection. Production control inspection, including sampling, shall conform to table I below and to the procedures for special inspection levels of MIL-STD-105. The inspection level shall be S-3 for normal, tightened, and reduced inspection. Production control inspection shall be performed on equipments that have passed production inspection. Production control tests shall be performed in accordance with MIL-E-16400. The equipment shall satisfactorily meet the requirements of production control inspection prior to release for shipment.

TABLE I. Production control inspection

Examination or test	Requirement paragraph	AQL (percent defective)
Weights and dimensions	3.5.1.4 3.5.1.5	10
Supply line voltage and frequency	3.4.3.1	10
Temperature	3.4.2	10
Power	3.5.2.2	10
Controls and control circuits	3.4.6	10
Enclosure	3.4.5.3	10

4.4.2.1 Rejected lots. If an inspection lot is rejected, the contractor may withdraw the lot from further inspection. The contractor may also rework a rejected lot to correct the defective units and reinspect the lot using tightened inspection. Rejected lots shall be kept separate from new lots and shall not lose their identity.

4.4.3 Environmental tests. Environmental tests shall be as listed in table II and shall be performed on units which have been subjected to and pass production inspection. Environmental tests shall be performed in accordance with MIL-E-16400.

TABLE II. Environmental tests

Examination or test	Requirement paragraph
Temperature	3.4.2
Humidity	3.4.2
Shock, vibration, inclination and noise	3.4.5.1, 3.4.5.2, 3.4.2(c)
Salt spray test	3.4.2

4.4.3.1 Environmental test samples. Samples for the environmental inspection shall be randomly selected in accordance with the following:

(a) One sample from the first 25 production units fabricated.
(b) One sample from each successive 75 units produced, or fraction thereof.

4.4.3.2 Nonconforming environmental sample units. If a sample unit fails the test specified in 4.4.3, the contractor shall immediately investigate the cause of failure and shall report to the Government inspector the results thereof and details of the corrective action taken to correct units of product which were manufactured with the same conditions, materials, processes; and so forth. If the Government inspector does not consider that the corrective action will enable the product to meet specified requirements, or if the contractor cannot determine the cause of failure, the matter shall be referred to the contracting officer.

4.4.3.3 Reinspection of conforming production control or environmental test units. Unless otherwise specified, sample units which have been subjected to and passed production control or environmental inspections may be accepted on the contract provided they are resubjected to and pass production inspection after repair of all visible damage.

4.5 Reliability Demonstration. The reliability of the suppression mixer shall be demonstrated and reported in accordance with MIL-STD-781.

4.5.1 Reliability qualification test. Reliability qualification testing shall be conducted on two first article equipments. Testing will be performed at Test Level C with Test Plan V for sufficient duration (in accordance with the requirements of MIL-STD-781) to permit either an accept or reject decision in accordance with the requirements of MIL-STD-781. The sample size for qualification testing shall be in accordance with table V of MIL-STD-781 for the initial production lot. The reliability requirements must be successfully demonstrated prior to the shipment of any production equipment.

Source: https://assist.dla.mil -- Downloaded: 2026-10-20T14:07Z
Check the source to verify that this is the current version before use.

4.5.2 Reliability sampling test. Reliability sampling testing shall be conducted on equipments selected from each three (3) month's production to ensure continued production of acceptable equipment. Sampling testing shall be performed at Test Level C with Test Plan XXV. Accept or reject criteria shall be in accordance with the parameters of Test Plan XXV summarized in table IV of MIL-STD-781. MIL-STD-781 shall be used to determine the number of units to be selected for reliability sampling tests.

4.5.3 Failure. Failure shall be defined as any malfunction or parameter deviation that prevents the equipment from performing within the operational requirements set forth in this specification.

4.5.4 Disposition of tested equipment. Upon completion of reliability testing, the tested equipment shall be re-inspected and conditioned as required, and may be delivered as part of the contract quantity, if it meets all terms and conditions of the contract for acceptance.

4.5.5 Rejected equipment. When equipment has been rejected as a result of the reliability test, the whole lot from which it was drawn shall be reworked as necessary, reinspected, and another random sample drawn for another reliability test. In the event that the cause of failure is equipment manufacturer design or part manufacturer design, corrective action shall be performed on all equipment delivered under the contract.

4.6 Maintainability demonstration. The maintainability requirements shall be successfully demonstrated in accordance with Method III of MIL-STD-471 prior to the shipment of any production equipment.

4.7 Inspection of preparation for delivery. Sample packages and packs and the inspection of the preservation and packaging, packing and marking for shipment and storage shall be in accordance with the requirements of section 5.

5. PREPARATION FOR DELIVERY

(The preparation for delivery requirements specified herein apply only for direct Government procurements. Preparation for delivery requirements or referenced documents listed in section 2 do not apply unless specifically stated in the contract or order. Preparation for delivery requirements for products procured by contractors shall be specified in the individual order.)

5.1 Preservation and packaging, packing, and marking. Unless otherwise specified, preparation for delivery shall be in accordance with MIL-E-17555 (see 6.2.1).

6. NOTES

6.1 Intended use. The equipment covered in this specification is intended for use in providing the necessary suppression gates to a single transponder from up to six suppression sources.

6.2 Ordering data. Procurement documents should specify the following:

6.2.1 Procurement requirements.

(a) Title, number, and date of this specification
(b) Number of first article samples to be submitted if other than specified in 4.3
(c) Marking, levels of preservation and packaging and packing required (see 5.1)

6.3 Definitions. For the purposes of this contract specification, the following terms are defined as stated:

Source: https://assist.dla.mil -- Downloaded: 2026-10-20T14:07Z
Check the source to verify that this is the current version before use.

6.3.1 Suppression source. A suppression source is defined as any electronic component, subsystem, or system which, because of potential over-the-air interference with the AIMS transponder, emits a signal intended to suppress the transponder's replies during the transmission time of the potentially interfering signal in space. Suppression sources fail into two categories as follows:

Category A: Short Pulse Source. A source whose suppression output signal is shorter in duration than its transmitted signal in space.

Category B: Long Pulse (Gate) Source. A source the duration of whose suppression output signal is equal to or greater than the duration of its transmitted signal in space.

6.3.2 Renewable Gate. A suppression pulse which, having been initiated by an input pulse, and receiving a second pulse during the gate's duration, automatically re-initiates its full duration starting from the leading edge of the second pulse, without losing amplitude. Two input pulses will be considered resolvable, in the sense of renewing the gate, if, between the trailing edge of the first pulse and the leading edge of the second pulse, the voltage at the input terminals drops to 2.7 volts or less.

6.4 First article. Invitations for bids should provide that the Government reserves the right to waive the requirement for first article samples as to those bidders offering a product which has been previously procured or tested by the Government, and that bidders offering such products, who wish to rely on such production or test, must furnish evidence with the bid that prior Government approval is presently appropriate for the pending procurement.

Preparing activity:
Navy - EC

(Project 5825-N006)

Source: https://assist.dla.mil -- Downloaded: 2026-10-20T14:07Z
Check the source to verify that this is the current version before use.

STANDARDIZATION DOCUMENT IMPROVEMENT PROPOSAL

OMB Approval
No. 22-R255

INSTRUCTIONS: The purpose of this form is to solicit beneficial comments which will help achieve procurement of suitable products at reasonable cost and minimum delay, or will otherwise enhance use of the document. DoD contractors, government activities, or manufacturers/ vendors who are prospective suppliers of the product are invited to submit comments to the government. Fold on lines on reverse side, staple in corner, and send to preparing activity. Comments submitted on this form do not constitute or imply authorization to waive any portion of the referenced document(s) or to amend contractual requirements. Attach any pertinent data which may be of use in improving this document. If there are additional papers, attach to form and place, both in an envelope addressed to preparing activity.

DOCUMENT IDENTIFIER AND TITLE

NAME OF ORGANIZATION AND ADDRESS

CONTRACT NUMBER

MATERIAL PROCURED UNDER A

☐ DIRECT GOVERNMENT CONTRACT ☐ SUBCONTRACT

1. HAS ANY PART OF THE DOCUMENT CREATED PROBLEMS OR REQUIRED INTERPRETATION IN PROCUREMENT USE?
   A. GIVE PARAGRAPH NUMBER AND WORDING.

   B. RECOMMENDATIONS FOR CORRECTING THE DEFICIENCIES

2. COMMENTS ON ANY DOCUMENT REQUIREMENT CONSIDERED TOO RIGID

3. IS THE DOCUMENT RESTRICTIVE?

   ☐ YES ☐ NO *(If "Yes", in what way?)*

4. REMARKS

SUBMITTED BY *(Printed or typed name and address - Optional)*

TELEPHONE NO.

DATE

DD FORM 1 JAN 72 1426 REPLACES EDITION OF 1 JAN 66 WHICH MAY BE USED

S/N 0102-014-1802

Source: https://assist.dla.mil -- Downloaded: 2026-10-20T14:07Z
Check the source to verify that this is the current version before use.